Exploratory Data Analysis — Report

Project 2 · Data Analytics Internship · Decode Labs

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Position:	Data Analytics Intern	**Mode:**	Remote / Virtual
Internship Period:	27 Sep 2026 – 27 Oct 2026	**Report Date:**	06 Oct 2026
Dataset:	Orders_Clean (cleaned in Project 1)	**Tool Used:**	Excel (PivotTables, formulas)

Objective

Analyze the cleaned Project 1 dataset to understand patterns, trends, and distributions, per the Project 2 brief: calculate basic statistics (mean, median, count), identify trends and outliers, and summarize key observations.

1. Descriptive Statistics

Metric	Quantity	UnitPrice	TotalPrice
Mean (Average)	2.95	356.41	1,053.97
Median	3.00	364.21	823.62
Count	1,200	1,200	1,200

The mean TotalPrice (1,053.97) sits approximately 28% above the median (823.62), indicating a right-skewed distribution: order values have a long tail of larger orders, so most orders fall below the average. Quantity and UnitPrice show mean and median close together, indicating a more balanced spread. Because TotalPrice is Quantity multiplied by UnitPrice, and both vary widely, a right-skewed total is consistent with that.

2. Outlier Analysis (IQR Method)

Metric	Value
Q1 (25th percentile)	410.52
Q3 (75th percentile)	1,578.48
IQR	1,167.96
Lower Limit	N/A (negative)
Upper Limit	3,330.41
Number of Outliers	8 (0.67% of dataset)

All 8 outliers in TotalPrice involve the maximum order quantity (5 units) of mid-to-high-priced products (Laptop, Printer, Tablet, Monitor, Chair). No data entry errors were found on inspection: these are legitimate bulk orders (signal, not noise), so no cleaning action was needed.

Do the outliers explain the mean-versus-median gap?

TotalPrice	Orders	Mean	Median	Mean above median
All orders	1,200	1,053.97	823.62	28.0%
Excluding the 8 outliers	1,192	1,038.36	816.11	27.2%

To test whether the 8 outliers cause the skew, the mean and median were recalculated without them. The gap barely moves (28.0% to 27.2%), so the outliers are the extreme end of a long tail but do not explain it. The right skew is a feature of the whole distribution.

3. Trend Analysis

By Product

Product	Order Count	Avg. TotalPrice
Chair	178	1,098.99
Desk	170	985.06
Laptop	173	1,110.56
Monitor	163	1,077.62
Phone	156	972.58
Printer	181	1,080.73
Tablet	179	1,042.28

Order volume is fairly even across products (156–181 orders). Phone has the lowest order count (156) and the lowest average order value (972.58), and Laptop the highest average order value (1,110.56), but these gaps are small and are not distinguishable from chance (see the significance checks below). The rankings also change when Cancelled and Returned orders are excluded.

By Referral Source

Referral Source	Order Count	Avg. TotalPrice
Email	250	1,047.23
Facebook	228	1,098.29
Google	241	1,039.18
Instagram	259	1,062.88
Referral	222	1,021.69

Order volume is fairly even across channels (222–259 orders). Facebook-referred orders have the highest average order value (1,098.29) and Instagram brings the most orders (259), but the gap between the highest and

lowest channel is only about 7%, and it is not distinguishable from chance. Excluding Cancelled and Returned orders, Google has the highest average order value (1,107.75). These figures do not support shifting marketing spend between channels.

Are the differences real?

Comparison (all 1,200 orders)	p-value
Average order value across products	0.62
Average order value across referral channels	0.89
Phone versus all other products (average order value)	0.16
Order counts across products	0.82
Order counts across referral channels	0.42

A p-value above 0.05 means gaps of this size could easily arise by chance, so none of these differences is established. This is not proof that no difference exists, only that this data gives no evidence of one. The tests (one-way ANOVA, and a t-test for Phone) were run on the raw dataset outside Excel. All comparisons count every order placed; 41.4% of orders (497 of 1,200) are Cancelled or Returned. Excluding them, Chair has the highest average order value (1,145.91) and Desk the lowest (961.56).

4. Correlation Analysis

Variable Pair	Correlation (r)
Quantity vs TotalPrice	0.615
Quantity vs ItemsInCart	0.650

Both pairs show a moderate positive correlation: related, but not strong enough to be treated as reliable predictors on their own. Other factors (such as UnitPrice variation across product categories) clearly also influence TotalPrice. Correlation is treated here as a clue for further investigation, not as evidence of a causal relationship.

5. Key Observations Summary

1. Order value is right-skewed: the mean (1,053.97) sits about 28% above the median (823.62), reflecting a long tail of larger orders across the whole distribution. The typical order is closer to the median than the average suggests.

2. 8 orders (0.67% of the dataset) exceed the IQR upper limit, all at the maximum quantity of 5 units. They are legitimate bulk orders, not errors, but they do not cause the skew: excluding them, the mean is still 27.2% above the median.

3. 41.4% of orders (497 of 1,200) are Cancelled or Returned. Every product and channel comparison in this report counts all orders placed, and some rankings change when those orders are excluded.

4. Phone has the lowest order count (156) and lowest average order value (972.58), and Laptop the highest average order value (1,110.56). The gaps are small and not distinguishable from chance ($p = 0.62$ across products; $p = 0.16$ for Phone versus the rest), so these are things to watch, not findings to act on.

5. Facebook-referred orders have the highest average order value (1,098.29), but channel differences are about 7% and not distinguishable from chance (p = 0.89), and Google leads once Cancelled and Returned orders are excluded. The data does not support shifting marketing spend.

6. Quantity shows a moderate (not strong) positive relationship with both TotalPrice and ItemsInCart: a clue for further analysis, not a standalone predictor.

Methodology

All analysis was performed in Microsoft Excel on the Orders_Clean dataset produced in Project 1. Descriptive statistics (mean, median, count) were calculated using AVERAGE, MEDIAN, and COUNT formulas on a dedicated Statistics sheet. Outliers were identified using the IQR method (1.5 times the IQR beyond Q1/Q3), then individually inspected to distinguish legitimate bulk orders (signal) from potential data errors (noise). The mean and median were then recalculated excluding the outliers to test whether they explain the skew. Trends were identified using PivotTables grouping order count and average order value by Product and by ReferralSource. Correlation was calculated using the CORREL function. Whether differences between products and channels exceed chance was checked with one-way ANOVA tests run on the raw dataset outside Excel. Every comparison counts all 1,200 orders placed, including Cancelled and Returned orders. Monetary values are shown as plain numbers because the dataset does not state a currency.

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